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Practical Work 4: Integration of observed data and example of parameter calibration

The objective of this practical session is to make you play the game of adapting a plant file to a new cultivar by using the parameter estimation tools available with STICS (either in JavaStics, or using the SticsRPacks packages, according to your wish).

We will use in this practical session a set of USMs and associated observations corresponding to field experiments of maize on different sites, treatments and years.

You will find in the **folder workspace_WP4** the STICS files corresponding to 7 USMs: sit1t096, sit1t196, sit1t296, sit1t396, sit396, sit2t197, sit2t397.
sit1t096 stands for: site number 1, treatment number 0 and year 1996.

There is therefore 5 USMs for the year 1996 on 2 different sites (4 treatments on site number 1, and 1 treatment on site number 3), and 2 USMs for year 1997, for site number 2.

Observed values of LAI (lai(n)), biomass (masec(n)), yield (mafruit), number of grains (chargefruit), starting date of filling of harvested organs (idrps) and maturity date (imats) are available for these USMs (see *.obs files). The number of observations per USM and variable is given in Table 1.

situation	chargefruit	idrps	imats	lai_n	mafruit	masec_n
sit1t096	1	1	1	11	4	5
sit1t196	1	1	1	11	4	6
sit1t296	1	0	0	11	3	5
sit1t396	1	0	0	11	3	5
sit2t197	1	1	1	14	3	3
sit2t397	1	1	1	14	3	3
sit396	1	1	1	17	5	6

Table 1. Number of observations per USM (column situation) and variable.

We do not have parameter values specifically adapted to the cultivar used in these experiments. This new cultivar is called "corntop". It is said to be quite similar to the "Furio" cultivar, for which we have parameter values for STICS, although it is supposed to have a lower yield.

TIP #1: read this document up to the end before starting to work 😉.
TIP #2: if you feel comfortable with R, we advise you to use SticsRPacks for this session. It offers more possibilities and flexibility than JavaStics for crop model calibration and evaluation. A script template is provided for this purpose (see file script_wp4.r).

I. Analyzing initial simulation results

In the plant file provided in workspace_WP4, we added the cultivar "corntop", and initialized the values of the corresponding parameters to these of the "Furio" cultivar. In the tec files associated with the given USMs, the cultivar "corntop" (#17) has been selected.

1. Run the simulations for all the given USMs

   JavaStics: menu "Running model → Run independent USMs"
   R: adapt the paths at the beginning of the script script_wp4.r and run it up to the first execution of the stics_wrapper function.

2. Compare observed and simulated data

   JavaStics: consider only the USMs s1t196, sit2197 and sit396 (for sake of brevity, their results being representative of the other USMs)
   - menu "Model outputs → Dynamic graphics"
   - Select the results to consider: button Select => choose the *.sti output file for a USM among the 3 listed above
   - Select the variables to plot: select both LAI "lai(n)" and Biomass "masec(n)" before clicking on the "Create chart" button to plot them on the same graph ; then do the same with the Yield "mafruit" and the Number of grains "chargefruit" ; and finally select separately the days of stage drp (idrps) and maturity (imats)
   - Click on the check box "Use observed values file from usm" so that observed values can be plotted
   - Click on the "View" button.
   - Keep the graph pop-up open, click on the "Reset selection" button and redo the procedure for the other 2 USMs.

   R: run part I.2 of the script and look at the generated plots and statistical criteria.

What can you say about the differences between observed and simulated values for the different variables?

II. Choosing the configuration of parameter estimation

Suppose your goal is to predict the yield of this new cultivar, as accurately as possible, using the STICS model:

- Which observed variable(s) will you use to achieve this goal?

- Given the list of cultivar parameters listed in the plant file, which parameter(s) do you think it might be relevant to modify?

- Which USM(s) will you use for the parameter estimation and which one(s) for their evaluation?

III. Estimating parameters' values for the new cultivar

Adapt the parameters' values of the cultivar "corntop" using observed values of simulated variables to improve yield prediction:

- Based on the answers you proposed in the last section, estimate the values of the parameters that seem relevant to you, from observations of a subset of variables and on the USMs you identified for the parameter estimation step.

  JavaStics:
  - menu "Running model → Run parameter optimization"
  - Select one by one the USMs to use for the parameter estimation (click on the USM and then on the "Add" button) and click on the "Next" button
  - Select the observed variable(s) to use and click on the "Next" button
  - Select the parameter(s) to estimate, and optionally its(their) bounds and initial values
  - Set the number of repetitions of the minimizations (each repetition will start from different initial values for the estimated parameters, that are randomly chosen if they are not given by the user)
  - Click on the "Run" button to start the parameter estimation.

  R:
  - fill in the blanks and run part II and III of the script
  - those who are comfortable with the SticsRPacks packages can try to provide parameter(s) value(s) adapted to the FURIO cultivar as initial values for the parameter estimation (see doc of estim_param function)
  - those who are very comfortable with this practical session (both R, R packages and parameter estimation concepts), and think they can complete before lunch, can try to estimate the parameter(s) on all the USMs and apply a leave-one-out procedure to evaluate the model prediction performances (warning: leave-one-out is not integrated into CroptimizR). The optimizations may however run during several hours in that case if you set xtol_rel=1e-3, thus try with xtol_rel=1e-2 to reduce computing time.

- Check the information given in output of the parameter estimation tool used. What can you say about the stability of the estimated values of the parameter(s) and minimized criterion? And about the relevance of the parameter(s) value(s) estimated? Are you confident with the results obtained at this step?

  JavaStics: look at the csv files generated, in particular the file "outputsOptim.csv" (open them in a text editor, e.g. Notepad++)

R: look at the pdf files generated.

IV. Evaluating the parameters' values proposed for the new cultivar

Assess the change in the model performances when simulating the cultivar corntop using the parameter values obtained in section II.

Did you reach your goal?

JavaStics:

- run the simulations on the USMs kept for evaluation using the original values of the parameters and plot the yield
- update the value(s) of the estimated parameter(s) for the cultivar corntop in the plant file "my_corn_plt.xml", run the simulations on the USMs kept for evaluation, plot the new values for the yield and check if it is better predicted.

R: run part IV of the script and check in the generated plots and goodness-of-fit criteria if yield is better simulated, particularly on the USMs kept for the evaluation.